

Villanova University

Department of Mechanical Engineering

ME 4850 SPRING 2026

**INTERIM REPORT 4. SYSTEM AND COMPONENT DESIGN OF A
POWER SYSTEM FOR A DEEP SPACE PROBE**

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1. Introduction and Background

This report was the first round of design for a deep-space probe power system. The goal was to design a system to harness energy from an isotope power source or a nuclear-powered reactor. The probe was intended to travel too far from the sun to be able to harness solar power. The system included a gas heater, a turbine, compressors, a heat-rejecting radiator, an intercooler, and a recuperator. Since this system used an all-gas flow, the system was thermodynamically modeled using a Brayton Cycle.

The main considerations of the design were to select a working fluid for the system, a mass flow rate, design the heat rejection radiator, and determine the maximum power and efficiency of the system. One important consideration in the system was the generation of entropy, which was the generation of losses in each component. To maximize power, efficiency, and probe lifetime, the losses needed to be minimized.

The main calculations of this project were conducted in the software Engineering Equation Software, or EES. This program was able to call material properties from a database to allow for easy calculations of the state points in the Brayton Cycle.

2. Design Specifications and Constraints

The probe was constrained to be only 2 meters in diameter, but one of the specifications was that the radiator expanded outward from the body of the probe. This impacted the design of the radiator, since the feasibility of deploying the chosen size must be accounted for, as well as how the working fluid was distributed to heat the entire surface.

Another important specification and assumption for this design was that the radiator and the heater were at constant pressure. This ensured that the only losses in those components were due to the entropy generation. The compressors and the turbine in the system operated due to pressure change, so those were not assumed to be isobaric.

The power generation was specified as 50kW and was assumed to be directed right to the gas heater. The efficiency of the heater was assumed to be 100%, with all of the heat being absorbed by the working fluid. The efficiency of a few other components were specified, along with other constraints and values, and are listed below in Table 1.

Table 1. Space Probe Specifications

Parameter	Value
Max Heat Input	50 kW
Turbine Efficiency	65%
Compressor Efficiency	75%
Heat Generator Efficiency	90%
Working Cycle	Brayton Cycle

Probe Shape	Sphere
Probe Size (diameter)	2 m
Surrounding Space Vacuum Temperature	3 K

3. Development of System Simulation Model

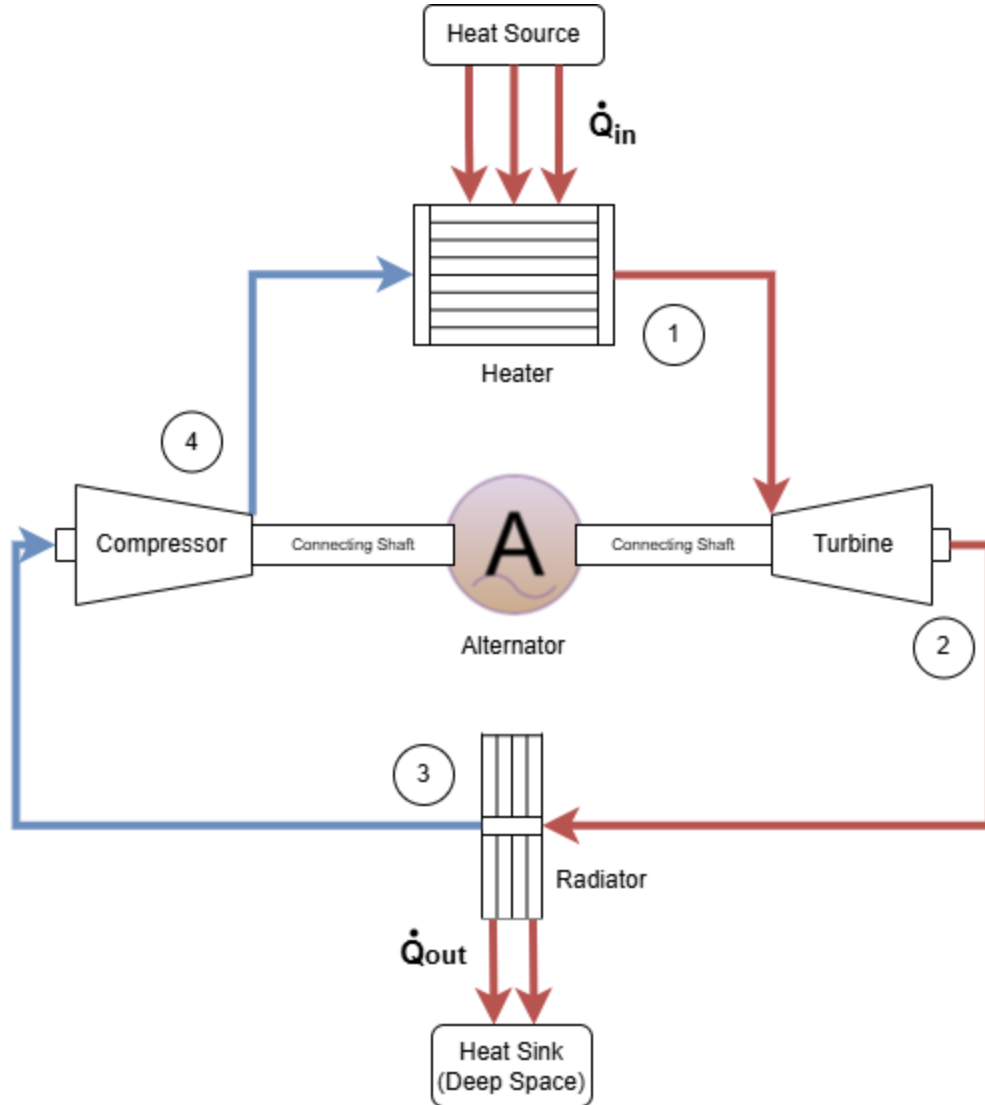


Figure 1. System Diagram

One of the main efficiency calculations for cycle systems was the Carnot efficiency. The principle of this equation, which is shown below in Equation 1, stated that the higher the temperature difference between the highest temperature (T_h) and the lowest temperature (T_L), the more efficient the system (η).

$$\eta = 1 - \frac{T_L}{T_H} \quad (1)$$

The maximum temperature occurred after the heater, where the energy entered the system. To maximize the temperature, the heater material had to be selected to withstand high temperatures while maintaining its strength. Research showed that titanium was an industry-standard material for similar applications, so titanium was selected as the gas heater material connected to the energy power source.¹ The chief engineer specified that a good maximum temperature estimate would be 70% of the titanium melting temperature, which was found to be around 1668°C.² The maximum temperature of the system was calculated to be 1150°C. The minimum temperature occurred at the surface of the radiator, since that was directly radiating with the cold vacuum of space. This temperature was calculated later by EES.

The four main components in this simplified Interim Report design were the heat source, the turbine, the radiator, and the compressors. For this report, the energy balance through the high-pressure and low-pressure compressors were assumed to be combined into a single compressor.

Each component was designed and considered by two different energy balance equations, based on the 1st and 2nd laws of thermodynamics. The first law of thermodynamics is stated below, showing that energy in must equal energy out. The energy was derived from the mass flow rate ($\dot{m}$), enthalpy (h), heat rate in ($\dot{Q}$), and work done by the system ($\dot{W}$).

$$(\dot{m} * h)_{in} + \dot{Q}_{in} = (\dot{m} * h)_{out} + \dot{W}_{out} \quad (2)$$

The second law of thermodynamics stated that there must be entropy conservation in a system. The conservation equation is shown below. The equation focused on the rate of entropy entering and leaving the system ($\dot{m} * s$), as well as the generation of entropy ($\dot{S}_{gen}$) and the entropy stored in the system ($\dot{S}_{sto}$).

$$\dot{S}_{gen} + (\dot{m} * s)_{in} = \dot{S}_{sto} + (\dot{m} * s)_{out} \quad (3)$$

The basic premise for determining each operating state was to set the initial temperature and pressure into the turbine, then let EES solve for the changes in enthalpy and entropy through each component. Using the entropy and enthalpy values and the designated pressure ratios and changes, the system temperature was determined for each state. To find the temperature, or anything related to a material's thermodynamic state, two thermodynamic properties had to be known.

The first component the working fluid went through was the turbine, shown below in Figure 2.

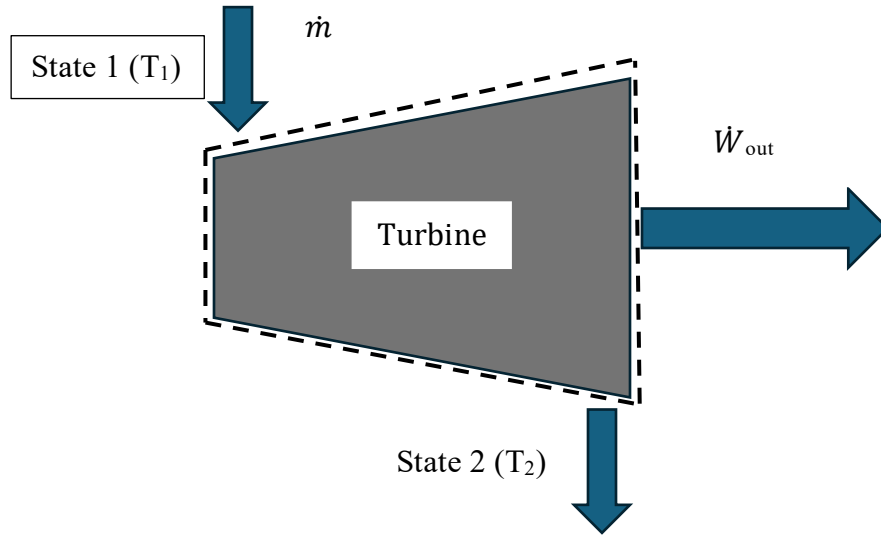


Figure 2. Turbine Component

The turbine was assumed to be isentropic, meaning there was no change in entropy through the component. The turbine took in the gas from state point 1 and returned the gas to the system at state point 2. The 1st Law energy balance for the turbine is shown below.

$$(\dot{m} * h)_1 = (\dot{m} * h)_2 + \dot{W}_{out} \quad (4)$$

$$\dot{W}_{out} = \dot{m} * (h_1 - h_2) \quad (5)$$

The 2nd Law entropy balance equation is shown below.

$$\dot{S}_{gen} + (\dot{m} * s)_1 = (\dot{m} * s)_2 \quad (6)$$

$$\dot{S}_{gen} = \dot{m} * (s_2 - s_1) \quad (7)$$

The amount of entropy generated by the turbine depended on the change in the entropy value of the carbon dioxide, which was later determined by EES. The computational solver was able to find the change in enthalpy across the turbine, and since the pressure values at the inlet and outlet were given design points, there were two thermodynamic values set and the outlet temperature could be found.

The next component in the system loop was the radiator, shown below in Figure 3.

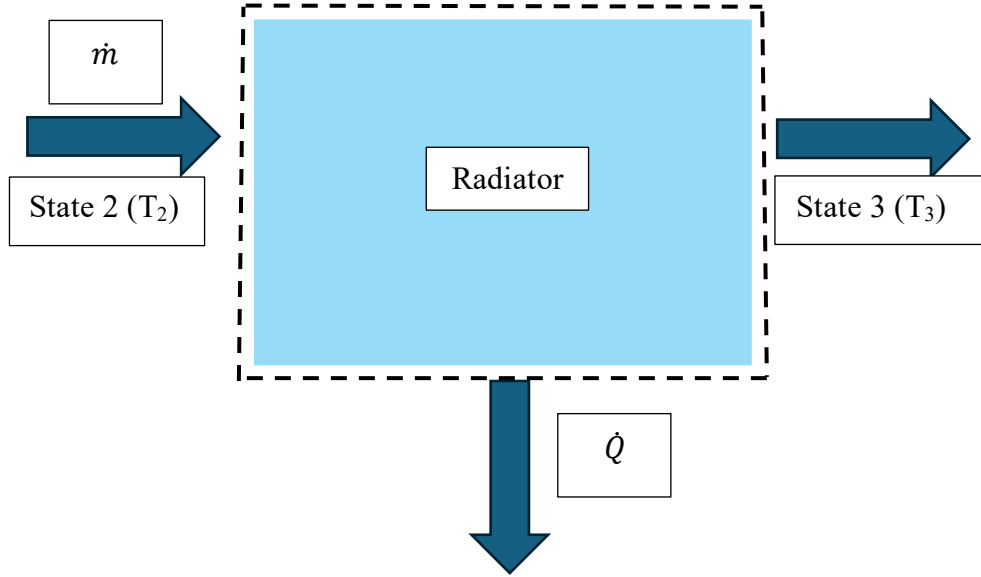


Figure 3. Radiator Component

The radiator was responsible for dissipating the heat from the system to ensure maximum temperature difference, and therefore maximum Carnot efficiency. The radiator was considered to be isobaric. The 1st Law energy balance for the radiator is shown below.

$$(\dot{m} * h)_2 = (\dot{m} * h)_3 + \dot{Q}_{out} \quad (8)$$

$$\dot{Q}_{out} = \dot{m} * (h_2 - h_3) \quad (9)$$

The 2nd Law entropy balance equation is shown below.

$$\dot{S}_{gen} + (\dot{m} * s)_2 = (\dot{m} * s)_3 \quad (10)$$

$$\dot{S}_{gen} = \dot{m} * (s_3 - s_2) \quad (11)$$

The amount of entropy generated by the radiator was dependent on the change in the entropy value of the carbon dioxide, which was later determined by EES.

The heat given off by the radiator was radiated into deep space. The radiation heat rate equation for this transfer accounted for the emissivity (ϵ), the Stefan-Boltzmann constant (σ), the total surface area of the radiator (A_s), the surface temperature of the radiator (T_s), and the temperature of space (T_o). The surface area of the radiator, as well as the material choice for the substrate and coating to determine the emissivity, will be designed in the Final Report.

$$\dot{Q}_{out} = \epsilon * \sigma * A_s * (T_s^4 - T_o^4) \quad (12)$$

The next component in the system was the compressor, modeled as a simplified combination of the high-pressure and low-pressure compressors. This was assumed for the Interim Report since the intercooler was not considered. The compressor is shown below in Figure 4.

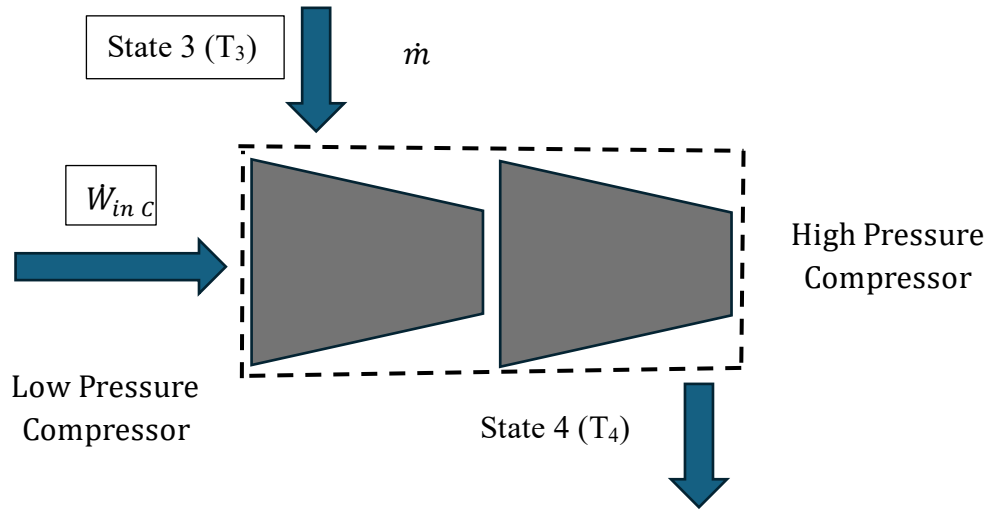


Figure 4. Compressor Component

The gas entered the low-pressure compressor first, then was fed directly into the high-pressure compressor, since this report did not consider an intercooler. The compressors were powered by the generated turbine work through the alternator. The alternator efficiency was specified as 90%; therefore, the work into the compressors was equal to 90% of the work leaving the turbine. The conversion is shown below.

$$\dot{W}_{in\ c} = 0.9 * \dot{W}_{out\ t} \quad (9)$$

The 1st Law energy balance for the compressor is shown below.

$$(\dot{m} * h)_3 + \dot{W}_{in\ c} = (\dot{m} * h)_4 \quad (10)$$

$$\dot{W}_{in\ c} = \dot{m}(h_4 - h_3) \quad (11)$$

The 2nd Law entropy balance equation is shown below.

$$\dot{S}_{gen} + (\dot{m} * s)_3 = (\dot{m} * s)_4 \quad (12)$$

$$\dot{S}_{gen} = \dot{m}(s_4 - s_3) \quad (13)$$

The amount of entropy generated by the compressor was dependent on the change in the entropy value of the carbon dioxide, which was later determined by EES. Since the combined compressor was modeled as adiabatic in this interim design, the entropy generation in this component was due to irreversibility associated with compression.

The final component in this simplified Brayton cycle system was the gas heater, which supplied the energy to the system by transferring the heat from the energy source to the working fluid. The gas heater component is shown in Figure 5.

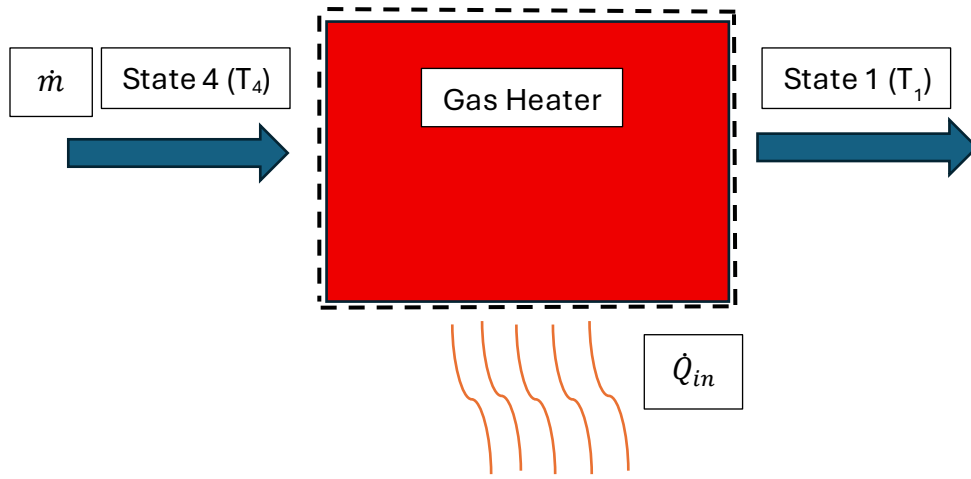


Figure 5. Gas Heater Component

The heat from the energy source passed into the heater and increased the temperature of the carbon dioxide. The heater was considered to be isobaric. The 1st Law energy balance for the gas heater is shown below.

$$(\dot{m} * h)_4 + \dot{Q}_l = (\dot{m} * h)_1 \quad (14)$$

$$\dot{Q}_l = \dot{m} * (h_1 - h_4) \quad (15)$$

The 2nd Law entropy balance equation is shown below.

$$S_{gen} + (\dot{m} * s)_4 + \frac{\dot{Q}_l}{T_H} = (\dot{m} * s)_1 \quad (16)$$

$$S_{gen} = \dot{m} * (s_1 - s_4) - \frac{\dot{Q}_l}{T_H} \quad (17)$$

The amount of entropy generated by the gas heater was dependent on the increase in the entropy of the carbon dioxide and the entropy transfer associated with heat addition from the source. Because the gas heater was treated as isobaric, the state at the turbine inlet could be determined directly from the specified heat input and the thermodynamic properties solved by EES.

4. EES Code Validation

There were several assumptions made in order to run the code. The first was the inlet temperature into the turbine, which was the maximum temperature of the system. This was assumed to be around 70% of the melting point of titanium and was input as 1300K. The other main input was the pressure at state one, entering the turbine. This was selected by looking at the critical point of the working fluid, CO₂. The critical point for CO₂ was at 32°C and around 7.4

MPa. At a pressure above this point, the fluid acted as a supercritical fluid. The selected pressure was chosen to be 1 MPa. In the final report, a full parametric analysis will be performed to select the ideal pressure to maximize system efficiency.

The code ran a total of 70 equations and 70 variables, which meant that it was able to solve for every operating point and property in the system [A-1]. The code report is shown below in Figure 6. Additionally, the P-h and T-s diagrams for the system using carbon dioxide are shown below in Figures 7 and 8. For this Interim Report, the EES code was run without any consideration of a recuperator or an intercooler between the high- and low-pressure compressors. The heat rejector was assumed to have an exiting temp of 350K, but further design will follow in the Final Report to be able to determine the actual temperature of the working fluid leaving the system.

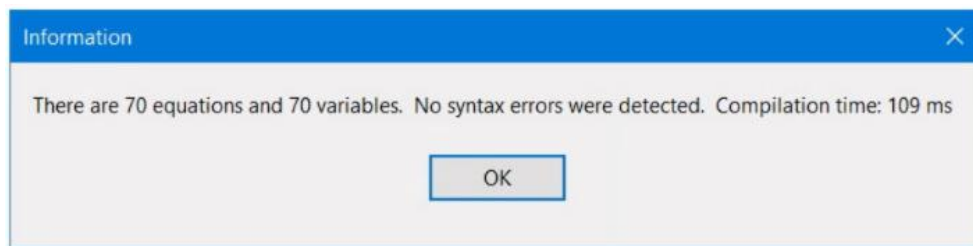


Figure 6. EES Code Check Window

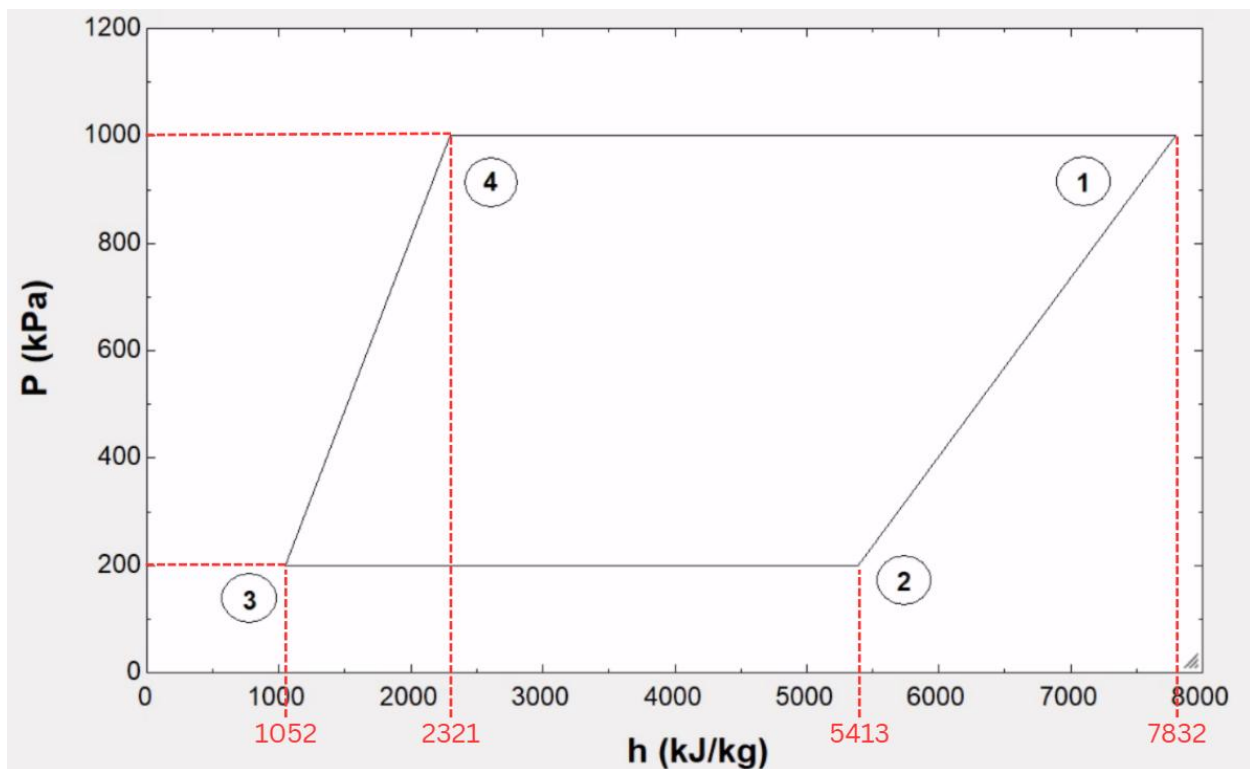


Figure 7. P-h Diagram for System without Recuperator or Intercooler

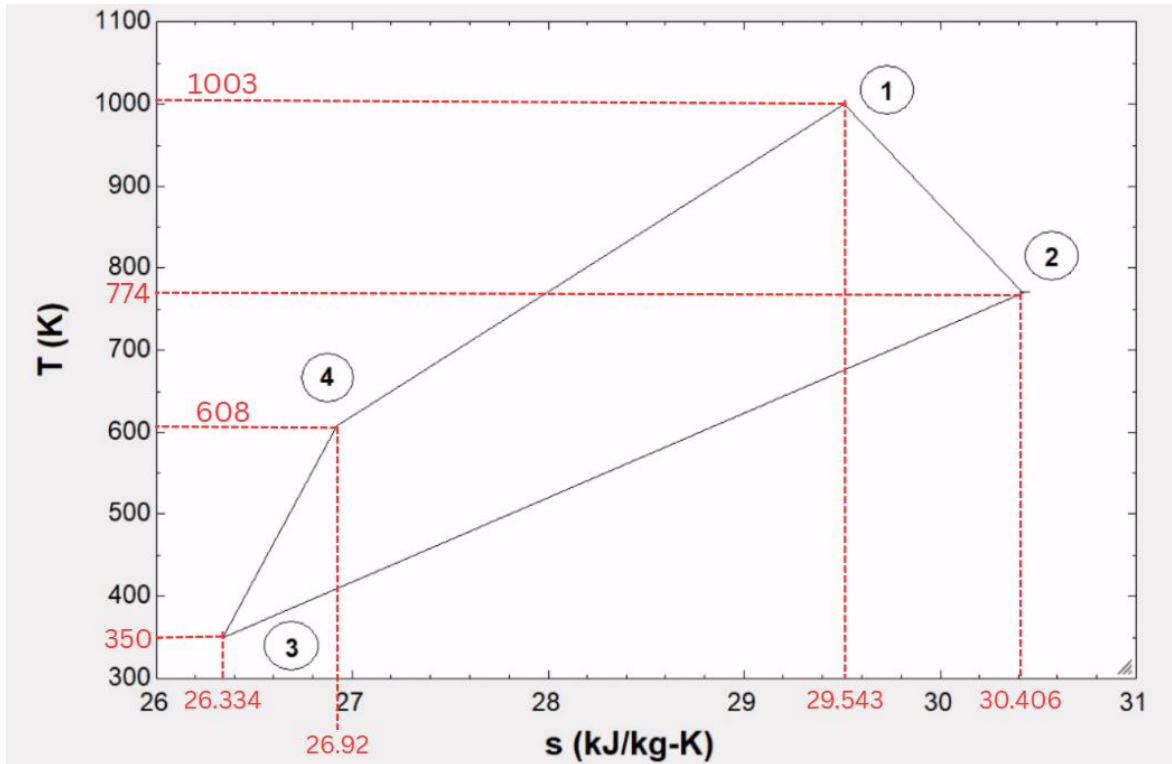


Figure 8. T-s Brayton Cycle Diagram for System without Recuperator or Intercooler

5. Operating States

The EES code solved for the four state points throughout the system by solving the input code consisting of 70 equations and 70 variables. The properties at the state points are shown below in Table 2. Initially there were problems with the maximum temperature of the CO₂ being set to 1100K, but when the pressure was increased to try and raise the maximum temperature, EES returned negative enthalpy values, which was deeply concerning. For the final report, a more optimal operating pressure and temperature, both in the supercritical region, will be chosen. For now, time and inexperience with the EES program were constraints on determining why an increase in pressure was resulting in such unnatural numbers. The operating values presented in the Table 2 show the calculated values for a 1000 kPa and 1100 K working fluid entering the turbine.

Table 2. State Point Operating Values

State Point	h (kJ/kg)	s (kJ/kg-K)	T (K)	P (kPa)
1	7832	29.543	1003	1000
2	5413	30.406	774	200
3	1052	26.334	350	200
4	2321	26.92	608	1000

6. Conceptual Design of Final Overall System

The system components that required design were the gas heater, the radiator, and the recuperator to be designed. All three make use of heat transfer to absorb, transfer, or dissipate heat from the working fluid, the carbon dioxide. For all designed components, material played a strong role in the decision making. Material affected the heat transfer throughout the components, but also the weight, which was important for any aerospace device.

The gas heater design needed to be highly effective to be able to absorb as much of the heat from the energy source as possible. This demonstrated a need for a large surface area of interaction between the incoming heat and the working fluid. This meant perhaps many tubes, or many passes of a few tubes, but a lot of surface area being the focus.

The radiator presented a different design challenge, since it was meant to be deployable outside the probe. Similar to the heater, the main priority was maximizing heat transfer. The idea for this design was to run the working fluid below the radiator surface in rectangular tubes and in a single tube with many passes. This would ensure maximum time and surface area for the working fluid to interact with the radiator surface. A crude SolidWorks model of this idea is shown below in Figure 10.

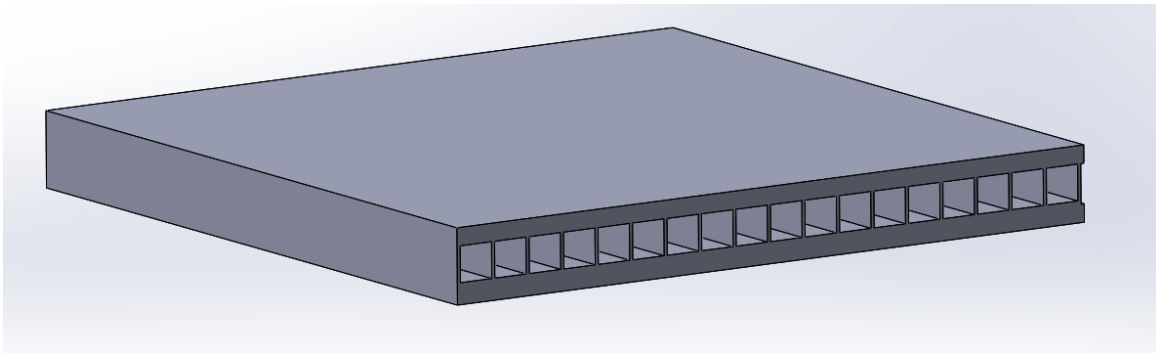


Figure 10. Conceptual Design of Radiator Component

Another consideration for the radiator was to utilize heat pipes to distribute the heat quicker and more efficiently throughout the system. For the Final Report, an investigation into the design and material needs of heat pipes, specifically oscillating heat pipes, will be conducted.

The recuperator component transferred heat from the working fluid leaving the turbine at state point 2 to working fluid going from the compressors to the gas heater at state point 4. The goal was to heat the working fluid entering the heater as much as possible to ensure that the heat transfer from the energy source can heat the fluid as high as possible. It also helps to conserve energy within the system and slightly cools the fluid at state 2 before it reaches the radiator for heat rejection. This design was theorized to use a counterflow heat exchanger system, maybe even a brazed plate heat exchanger to minimize overall size while maximizing contact between the hot and cold working fluid.

The other design points in the system included choosing a mass flow rate, a pressure ratio in the turbine, and a maximum temperature of the fluid entering the system. In the Final Report, a parametric study will be conducted to determine and analyze the effects of changing each of these variables to see how it would affect entropy generation throughout the system as well as overall efficiency. In the Interim code, the pressure ratio heavily affected how much work was done by the turbine, which impacted the ability of the compressor system as well to increase the temperature of the fluid after it had gone through the radiator. The EES coding will be vital to discern the interconnected relationships of all of the variables, because one change snowballs into affecting all of the other components' operating states.

7. Future Work

The Final Report will include a full design of the radiator, the gas heater, the compression system (with and without intercooling), and a full design of the system operating parameters, such as operating pressure and temperature entering the turbine, as well as a finalized operating fluid. For this report, carbon dioxide was chosen as the operating fluid, but it was found to have a temperature limitation on EES of 1100K. Other options for a working fluid include helium, which had no temperature limit, and would be able to model the maximum temperature as determined by the material of the gas heater.

The radiator will be designed to maximize heat transfer from the working fluid to the space vacuum, with the goal of maximizing the Carnot efficiency as discussed in Section 3. The recuperator will be designed to maximize heat transfer between the working fluids at states 2 and 4. The new part of the Final Report will be to investigate the pros and cons of an intercooler between the high- and low-pressure compressors. The theoretical motivation for including an intercooler was to cool the working fluid before entering the second stage of compression to reduce the operating power needed.

There will also be an investigation into the production of electrical energy by the alternator between the turbine and the compressors. The compressor pressure ratios also need to be determined. Once all of these factors are designed and written into EES, the final operating states can be determined, and the overall system efficiency will be calculated.

References

- (1) *Best Metals for High Temperature Applications* | *Metal Supermarkets*.
<https://www.metalsupermarkets.com/best-metals-high-temperature-applications/> (accessed 2026-03-30).
- (2) *Melting Point of Common Metals, Alloys, & Other Materials* | *AMERICAN ELEMENTS*®.
<https://www.americanelements.com/meltingpoint.html> (accessed 2026-03-30).

Appendix

A. Engineering Equation Solver

```
Q_dot_in = 50000 [W] "Max heat input"
eta_t = 0.65 "Turbine efficiency"
eta_c = 0.75 "Compressor efficiency"
eta_alt = 0.9 "Generator efficiency"
T_sink = 3 [K] "Deep sink temperature"

T_1 = 1100 [K] "Turbine inlet temp"
P_1 = 2200 [kPa] "Turbine inlet pressure"
PR = 5.0 "Pressure ratio P1/P2"
m_dot = Q_dot_in / (h_1 - h_8) "mass flow rate calculation"

h_1 = Enthalpy(CarbonDioxide, T=T_1, P=P_1) "Turbine enthalpy"
s_1 = Entropy(CarbonDioxide, T=T_1, P=P_1) "Turbine entropy"

P_2 = P_1 / PR "Turbine exit pressure"

s_2s = s_1 "Isentropic turbine exit entropy"
h_2s = Enthalpy(CarbonDioxide, P=P_2, s=s_2s) "Turbine exit enthalpy"

h_2 = h_1 - eta_t * (h_1 - h_2s) "Turbine exit enthalpy from efficiency"

T_2 = Temperature(CarbonDioxide, P=P_2, h=h_2) "Turbine exit temperature"
s_2 = Entropy(CarbonDioxide, P=P_2, h=h_2) "Turbine exit entropy"

T_3 = T_2 "Radiator inlet temperature"
P_3 = P_2 "Radiator inlet pressure"
h_3 = h_2 "Radiator inlet enthalpy"
s_3 = s_2 "Radiator inlet entropy"

T_4 = 250 [K] "Radiator outlet temperature"
P_4 = P_2 "Radiator outlet pressure"
h_4 = Enthalpy(CarbonDioxide, T=T_4, P=P_4) "Radiator outlet enthalpy"
s_4 = Entropy(CarbonDioxide, T=T_4, P=P_4) "Radiator outlet entropy"

q_dot_rad = m_dot * (h_3 - h_4) "radiator heat rejection"

s_7s = s_4 "Compressor inlet entropy"
h_7s = Enthalpy(CarbonDioxide, P=P_1, s=s_7s) "Compressor inlet enthalpy"
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h_7 = h_4 + (h_7s - h_4) / eta_c "Compressor exit enthalpy"

T_7 = Temperature(CarbonDioxide, P=P_1, h=h_7) "Compressor exit temperature"
s_7 = Entropy(CarbonDioxide, T=T_7, P=P_1) "Compressor exit entropy"

T_8 = T_7 "Heater inlet temperature"
P_8 = P_1 "Heater inlet pressure"
h_8 = h_7 "Heater inlet enthalpy"
s_8 = s_7 "Heater inlet entropy"

Q_check = m_dot * (h_1-h_8) "Heater energy balance"

W_dot_t = m_dot * (h_1 - h_2)
W_dot_c = m_dot * (h_7 - h_4)
W_dot_net = W_dot_t - W_dot_c
W_dot_elec = eta_alt * W_dot_net

BWR = W_dot_c / W_dot_t "Back work ratio"
eta_sys = W_dot_elec / Q_dot_in "System efficiency"

|
T[1] = T_1 "State points for T-s and P-h diagrams"
s[1] = s_1
h[1] = h_1
P[1] = P_1
T[2] = T_2
s[2] = s_2
h[2] = h_2
P[2] = P_2
T[3] = T_3
s[3] = s_3
h[3] = h_3
P[3] = P_3
T[4] = T_4
s[4] = s_4
h[4] = h_4
P[4] = P_4
T[5] = T_7
s[5] = s_7
h[5] = h_7

```

Figure A-1. EES Code